Immediate Learning Outcomes of AI- and Sensor-integrated Versus Face-to-face CPR Retraining for Chest Compression Quality: A Randomized Trial

Abstract

This study aims to evaluate the immediate learning outcomes of an AI- and sensor-integrated cardiopulmonary resuscitation (CPR) retraining system (Intelligent Practical Training System, IPTS) with conventional face-to-face retraining in healthcare professionals. We conducted a randomized trial of 136 healthcare professionals, allocated 1:1 to a 30-minute IPTS or face-to-face groups. IPTS integrates AI and sensors to enable instructor-free learning and examiner-free assessment. Learning outcomes were measured objectively using the manikin, without providing any feedback or assessment results to participants. Primary outcomes were chest compression depth (CC) accuracy and CC rate. Secondary outcomes included tidal volumes of the two rescue breaths and CPR self-efficacy. No evidence supports the superiority of IPTS in CC depth accuracy ($P_{\text{adj}} = .878$) or CC rate ($P_{\text{adj}} = .729$). However, the IPTS group achieved significantly superior rescue breath quality (first: rank-biserial r=-.585, 95% CI [-0.737, -0.416], $P_{\text{adj}} < .006$, large; second: rank-biserial r=-.512, 95% CI [-0.687, -0.323], $P_{\text{adj}} < .006$, large), with median values falling within the American Heart Association recommended range (first: 548.99 mL, second: 542.68 mL), whereas the face-to-face group's values were excessively high (first: 806.97 mL, second: 800.23 mL). Both groups showed significant and similar improvements in CPR self-efficacy, indicating comparable psychological benefits. Under controlled conditions, IPTS did not demonstrate superiority in immediate outcomes for CC quality and CPR self-efficacy compared to sensor-free face-to-face instruction, but achieved superior immediate outcomes for rescue breaths. These results demonstrate the feasibility of a novel integration of AI and sensors in medical education.